Supplementary Parameter Ledger

Robust Closed-Loop Restoration for Mobility-Coupled Distribution Systems with Measured Charging and Executable Actions

Scope. This ledger reports the implemented value, unit, role, provenance or calibration rule, selection data, and challenged range for every manuscript coefficient and major execution setting. Dimensionless objective weights define a benchmark score, not monetary cost. The CSV source is `config/full_parameter_ledger.csv`.

Table S1: Complete parameter and hyperparameter ledger.

Group	Symbol	Value	Unit	Role	Source / calibration	Selection data	Sensitivity / range
forecast	L	12	h	observed history length	fixed training configuration	2018-2022 Boulder EV data	not varied
forecast	H	6	h	rollout and forecast horizon	fixed training configuration	2018-2022 Boulder EV data	not varied
forecast	hidden	32	units	graph-recurrent hidden width	validation configuration	2022 Boulder validation	not varied
forecast	epochs	40	epochs	maximum training epochs	validation configuration	2022 Boulder validation	not varied
forecast	batch	256	samples	training batch size	validation configuration	2022 Boulder validation	not varied
forecast	learning_rate	0.002	none	Adam learning rate	validation configuration	2022 Boulder validation	not varied
forecast	weight_decay	0.0001	none	L2 regularization	validation configuration	2022 Boulder validation	not varied
forecast	dropout	0.1	fraction	dropout probability	validation configuration	2022 Boulder validation	not varied
forecast	physics_lambda	0.08	none	physics-guidance training weight	retained training configuration	2022 Boulder validation	plain inference mode reported
forecast	smooth_lambda	0.002	none	temporal smoothness weight	retained training configuration	2022 Boulder validation	plain inference mode reported
objective	gamma	1.0	none	finite-horizon discount factor	undiscounted implementation	all scenarios	not varied
objective	lambda_m	1.8	relative weight	mobility-delay contribution	declared benchmark weight	fixed before paired evaluation	3.6 in objective sensitivity
objective	lambda_e	2.6	relative weight	unserved-energy contribution	declared benchmark weight	fixed before paired evaluation	5.2 in objective sensitivity
objective	lambda_c	1.7	relative weight	communication-loss contribution	declared benchmark weight	fixed before paired evaluation	3.4 in objective sensitivity
objective	lambda_p	1.2	relative weight	power-service contribution	declared benchmark weight	fixed before paired evaluation	2.4 in objective sensitivity
objective	lambda_z	12.0	relative weight	cross-domain exposure contribution	declared benchmark weight	fixed before paired evaluation	6.0 in objective sensitivity
service	beta_m	0.13	relative load	mobility contribution to communication load	declared benchmark assumption	fixed before paired evaluation	not varied
service	beta_e	0.19	relative load	energy contribution to communication load	declared benchmark assumption	fixed before paired evaluation	not varied
service	beta_r	6.0	relative load	road-threat contribution to communication load	declared benchmark assumption	fixed before paired evaluation	not varied
service	beta_p	3.5	relative load	power-threat contribution to communication load	declared benchmark assumption	fixed before paired evaluation	not varied
service	alpha_c	0.40	none	direct communication-threat derating	declared benchmark assumption	fixed before paired evaluation	not varied
service	alpha_p	1.0	none	power-availability derating in service equation	normalized threat-state definition	fixed before paired evaluation	not varied
service	rho_c	0.45	none	communication shortfall to mobility delay	declared benchmark assumption	fixed before paired evaluation	not varied
service	rho_q	0.22	none	backlog carryover	declared benchmark assumption	fixed before paired evaluation	requires less than 1
allocator	kappa_q	1.0	relative weight	backlog in charging score	declared benchmark assumption	fixed before paired evaluation	not varied
allocator	kappa_ch	1.5	relative weight	current threat in charging score	declared benchmark assumption	fixed before paired evaluation	not varied
allocator	kappa_E	0.7	relative weight	energy in communication score	declared benchmark assumption	fixed before paired evaluation	not varied
allocator	kappa_com	1.5	relative weight	current threat in communication score	declared benchmark assumption	fixed before paired evaluation	not varied
allocator	omega_p	1.6	relative weight	power threat in repair priority	declared benchmark assumption	fixed before paired evaluation	not varied
allocator	omega_c	1.3	relative weight	communication threat in repair priority	declared benchmark assumption	fixed before paired evaluation	not varied
allocator	omega_r	1.1	relative weight	road threat in repair priority	declared benchmark assumption	fixed before paired evaluation	not varied
allocator	omega_Gamma	0.8	relative weight	multi-step marginal benefit	declared benchmark assumption	fixed before paired evaluation	not varied
allocator	eta_service	0.04	fraction	uniform reserve for charging and communication	declared benchmark assumption	fixed before paired evaluation	not varied
allocator	eta_repair	0.02	fraction	uniform reserve for repair priority	declared benchmark assumption	fixed before paired evaluation	not varied
allocator	epsilon	1e-9	score unit	square-root stabilizer	numerical implementation	fixed	positive by construction
budget	B_ch	29.433750	kWh per h	total hourly charging allocation	1.22 times all-station historical mean hourly energy	all 2018-2023 Boulder sessions	stress uses 24.4 factor with 20 times demand
budget	B_com	73.144081	service unit	total communication allocation	0.72 times supplied station capacity proxy	processed Boulder benchmark	not varied
budget	B_res	10.0	priority unit	total requested restoration priority	max of 10 and 0.018 times mean arrivals	processed Boulder benchmark	not varied
transition	rho_r	0.72842984	none	road-state persistence	nonnegative deficit regression	2018-2021 Boulder station-hours	0.54632238 to 0.91053730
transition	sigma	0.23278108	none	spatial propagation	nonnegative deficit regression	2018-2021 Boulder station graph	0.11639054 to 0.34917162
transition	rho_p	0.77180562	none	power-state persistence	day-block bootstrap	aligned EAGLE-I Boulder observations	0.75369266 to 0.78666740
transition	tau_pr	0.0	none	power-to-mobility association	nonnegative observational regression	aligned EAGLE-I and Boulder EV deficits	0.0 to 0.03180810
transition	tau_pc	0.16645075	none	power-to-communication coupling	finite-buffer packet stress construction	packet scenarios	0.01178179 to 0.27083775
transition	rho_c_state	0.5	none	communication-state persistence	sensitivity-only envelope	no longitudinal field packet trace	0.3 to 0.8
transition	tau_rc	0.04	none	road-to-communication coupling	sign-constrained envelope	no joint field trace	0.0 to 0.08
transition	tau_cr	0.08	none	communication-to-road coupling	sign-constrained envelope	no joint field trace	0.0 to 0.16
transition	tau_cp	0.04	none	communication-to-power-service coupling	sign-constrained envelope	no joint field trace	0.0 to 0.08

Continued on next page

Table S1 continued

Group	Symbol	Value	Unit	Role	Source / calibration	Selection data	Sensitivity / range
repair	road_effect	0.08	normalized state	predictive priority quantum effect	declared benchmark assumption	fixed before paired evaluation	continuous prediction layer only
repair	power_effect	0.12	normalized state	predictive priority quantum effect	declared benchmark assumption	fixed before paired evaluation	continuous prediction layer only
repair	communication_effect	0.10	normalized state	predictive priority quantum effect	declared benchmark assumption	fixed before paired evaluation	continuous prediction layer only
crew	vehicles	4	integer	available repair crews	declared benchmark setting	all integrated scenarios	12 and 24 in sensitivity
crew	jobs	36	integer	requested outage work orders per scenario	declared benchmark setting	all integrated scenarios	not varied
crew	speed	30	km per h	road travel speed	declared benchmark setting	Boulder station coordinates	not varied
crew	circuity	1.25	ratio	road-to-great-circle distance multiplier	declared benchmark setting	Boulder station coordinates	not varied
crew	service_time	empirical	h	job service duration	EAGLE-I 50-customer post-peak half-recovery samples	497 observable events clipped to 0.25-8 h	multiplied by 0.5 1.0 and 2.0 in routing study
crew	solver	deterministic list	none	priority per incremental completion time with stable tie-break	implementation rule	common route portfolio	no wall-clock cutoff
packet	duration	120	s	queue simulation window	declared stress setting	20 selected 2023 hours	not varied
packet	arrival_rate	8+0.8 activity	packet per s	station traffic arrival intensity	declared stress function	observed connected sessions and arrivals	traffic multipliers 0.5 1 2 and 4
packet	service_rate	30	packet per s	nominal queue service rate	declared stress setting	packet stress simulator	power and communication threat derate it
packet	buffer	100	packets	finite FCFS buffer	declared stress setting	packet stress simulator	not varied
packet	control_fraction	0.05	fraction	share of arrivals carrying control	declared stress setting	packet stress simulator	not varied
packet	deadline	500	ms	on-time control deadline	declared stress setting	packet stress simulator	not varied
packet	max_retries	2	integer	retransmission limit	declared stress setting	packet stress simulator	not varied
packet	backoff_mean	0.05	s	exponential retry backoff mean	declared stress setting	packet stress simulator	not varied
packet	erasure	min(0.25;0.002+0.08 z_c)	probability	threat-dependent packet erasure	declared stress function	packet stress simulator	not varied
packet	backup	60	s	primary backup duration	declared stress setting	packet stress simulator	0 and 300 s comparisons
grid	feeder	SMART-DS SFO P1U	case	independent AC execution model	public SMART-DS v1.0 complete feeder	selected before policy comparison	one feeder and 20 mappings
grid	power_factor	0.98	lagging	EV reactive-power assumption	declared execution setting	all online OpenDSS actions	not varied
grid	load_model	constant-PQ model 1	OpenDSS model	balanced three-phase wye EV increment	released eligible load connection and model	17 three-phase loads	not varied
grid	voltage_band	0.95-1.05	p.u.	accepted active-node voltage range	declared operational gate	all online OpenDSS actions	not varied
grid	line_limit	1.0	p.u.	accepted maximum branch loading	declared operational gate	all online OpenDSS actions	not varied
grid	bisection_iterations	14	integer	common-fraction AC projection tolerance	implementation setting	all infeasible raw actions	fraction resolution about 6.1e-5
grid	mappings	20	integer	seeded station-to-load embeddings	geographic rank and seeded assignment	Boulder stations to 17 eligible SMART-DS loads	scenario id modulo 20
grid	mapping_seed	20260825	integer	station mapping random seed	prespecified seed	all mappings	not varied
station_flow	capacity	max(q0.995*1.2;7.2)	kWh per h	station service-capacity stress proxy	training-period empirical quantile	2018-2021 Boulder energy	not a charger nameplate claim
station_flow	closure_fraction	0.1 0.2 0.3	fraction	closed-station stress	declared validation grid	24 selected 2023 hours	full factorial
station_flow	demand_multiplier	1 2 4	ratio	demand stress	declared validation grid	24 selected 2023 hours	full factorial
station_flow	distance_beta	0.5 1 2	per km	routing distance penalty	declared validation grid	24 selected 2023 hours	full factorial
station_flow	loyalty	2.0	cost unit	penalty for changing station	declared validation setting	24 selected 2023 hours	not varied
statistics	n_primary	4096	paired scenarios	primary scenario count	prespecified four equal groups	seed 2026	1024 per threat class
statistics	bootstrap	20000	resamples	paired mean-difference interval	prespecified inference	primary paired scenarios	percentile 95 percent interval
statistics	holm_family	5	tests	total-cost primary family	prespecified multiplicity family	overall plus four threat classes	paired t and Wilcoxon adjusted separately

Execution note. Continuous restoration values are requested priorities used for candidate scoring. They do not reduce the realized threat state. Packet delivery determines dispatched work orders; the deterministic integer route records completion events; only those events clear the corresponding realized threat state.